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Transcriptional Responses of *Mycobacterium tuberculosis* to Hypoxia, Rifampicin Exposure and Iron Starvation

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Abstract

Mycobacterium tuberculosis (Mtb) adapts to hostile host environments through extensive transcriptional reprogramming. Environmental stresses such as hypoxia, antibiotic exposure, and nutrient deprivation induce distinct physiological states that enable bacterial persistence and survival. Understanding these transcriptional adaptations is essential for identifying conserved stress-response mechanisms and condition-specific regulatory pathways. The objective of this study was to develop a reproducible computational framework for analyzing transcriptomic responses of Mtb H37RV strain across three biologically distinct stress conditions: Hypoxia, Rifampicin exposure and Iron starvation. Differential expression analysis was performed independently within each experimental study, followed by cross-condition comparison to identify shared and unique stress-responsive genes.

RNA-seq count matrices generated by featureCounts were analyzed using DESeq2. Quality assessment was performed through variance stabilizing transformation (VST) and principal component analysis (PCA). Differential expression analysis was conducted independently for each condition using shrinkage estimation of log2 fold changes with apeglm. Significant genes were identified using adjusted p-value <0.05 and $|\log_2FC| > 1$. Downstream analyses included volcano plots with significant genes per condition, Venn diagrams and concordance analysis of common genes and KEGG enrichment.

The framework provides a reproducible strategy for comparative transcriptomic analysis of Mtb H37RV strain in different functional states while avoiding incorrect statistical modeling caused by study-condition confounding.

1. Introduction

Tuberculosis (TB) remains one of the most devastating infectious diseases worldwide and continues to pose a major global public health challenge despite decades of intensive research and the availability of effective chemotherapy. The disease is caused by *Mycobacterium tuberculosis* (Mtb), a slow-growing intracellular pathogen that primarily infects the lungs but can disseminate to almost every organ in the human body. According to the World Health Organization (WHO), tuberculosis remains among the leading causes of death from a single infectious agent, causing millions of new infections and over one million deaths annually. The

continued emergence of multidrug-resistant (MDR-TB) and extensively drug-resistant tuberculosis (XDR-TB), together with the persistence of latent tuberculosis infections, highlights the urgent need to better understand the biology of this pathogen and identify novel therapeutic targets.

One of the most remarkable characteristics of *M. tuberculosis* is its extraordinary ability to survive under a wide range of hostile environmental conditions encountered within the host. Following infection, the bacterium is exposed to numerous physiological stresses imposed by the host immune system, including oxygen limitation, nutrient deprivation, acidic pH, oxidative stress, nitrosative stress, and prolonged exposure to antimicrobial drugs. *M. tuberculosis* responds by undergoing extensive physiological and metabolic reprogramming that enables it to adapt, persist, and evade host defenses. These adaptive responses contribute significantly to disease progression, latent infection, treatment failure, and relapse.

Among the numerous environmental stresses encountered by *M. tuberculosis*, hypoxia, antibiotic exposure, and iron limitation represent three biologically significant conditions that profoundly influence bacterial physiology.

Hypoxia is one of the defining characteristics of granulomatous lesions formed during latent tuberculosis infection. Within granulomas, oxygen availability decreases substantially, forcing *M. tuberculosis* to shift from active replication toward a non-replicating persistent state. This transition is mediated by large-scale transcriptional reprogramming, most notably through activation of the DosR (Dormancy Survival Regulator) regulon, which controls genes involved in anaerobic metabolism, stress adaptation, and long-term survival. Understanding hypoxia-induced transcriptional responses is therefore critical for studying latent tuberculosis and bacterial persistence.

Exposure to antibiotics, particularly rifampicin, represents another major environmental challenge for *M. tuberculosis*. Rifampicin is one of the most important first-line anti-tuberculosis drugs and exerts its antibacterial activity by inhibiting the β -subunit of DNA-dependent RNA polymerase, thereby preventing transcription. Although rifampicin is highly effective against actively growing bacteria, *M. tuberculosis* can undergo substantial transcriptional remodeling during drug exposure, activating stress-response pathways, metabolic adaptation mechanisms, and survival programs that contribute to drug tolerance. Characterizing these transcriptional responses provides important insights into mechanisms of antimicrobial resistance and persistence.

Iron starvation constitutes another physiologically relevant stress encountered by *M. tuberculosis* within the host. Iron is an essential micronutrient required for numerous enzymatic reactions, electron transport processes, DNA synthesis, and cellular respiration. However, the host immune system restricts iron availability through a defense mechanism known as nutritional immunity, limiting bacterial access to this critical resource. In response, *M. tuberculosis* activates iron

acquisition systems, siderophore biosynthesis pathways, and extensive metabolic adaptations to maintain intracellular iron homeostasis. Studying these transcriptional changes provides valuable information regarding bacterial survival strategies during nutrient limitation.

Although numerous transcriptomic studies have investigated individual stress conditions, most analyses are performed independently and focus on a single environmental challenge. Consequently, relatively little is known about the extent to which transcriptional responses are conserved across different stress conditions or whether each environmental stress induces unique regulatory programs. Comparative transcriptomic analysis offers an opportunity to distinguish universal bacterial stress-response mechanisms from condition-specific adaptations, thereby improving our understanding of *M. tuberculosis* physiology.

However, comparative analysis of publicly available RNA-seq datasets presents important statistical challenges. Public datasets are typically generated by different laboratories using distinct experimental designs, sequencing platforms, library preparation protocols, growth conditions, and sample processing procedures. These technical differences introduce study-specific batch effects that can strongly influence gene expression measurements. A particularly challenging situation arises when each biological condition originates from a separate independent study, resulting in complete confounding between biological condition and study. Under such circumstances, any attempt to perform pooled differential expression analysis across all samples would produce results that cannot distinguish genuine biological differences from study-specific technical variation. Consequently, pooled analysis would yield statistically uninterpretable estimates of differential expression.

To address this limitation, the present study adopts a statistically rigorous analysis strategy in which differential expression analysis is performed independently within each experimental study. Each stress condition is analyzed separately using matched treatment and control samples originating from the same experiment, thereby eliminating confounding between biological condition and study. Differentially expressed genes identified within each condition are subsequently compared across studies to identify conserved and condition-specific transcriptional responses without violating the assumptions underlying differential expression analysis.

In this study, a reproducible computational framework was developed to investigate pathogen-side transcriptomic responses of *Mycobacterium tuberculosis* under three distinct stress conditions: hypoxia, rifampicin exposure, and iron starvation. RNA-seq count data generated from independent public studies were analyzed using the DESeq2 statistical framework. Differential expression analysis was conducted independently for each condition to avoid study-related confounding, followed by visualization of transcriptional responses using principal component analysis and volcano plots. Significant differentially expressed genes were subsequently compared across conditions to identify shared and unique stress-response

signatures. Functional interpretation was performed through KEGG pathway enrichment analysis.

2. Materials and Methods

2.1 Experimental design and confounding structure

The dataset comprises 18 samples across three independent studies, one per condition (hypoxia, rifampicin, iron starvation), each contributing 3 treatment and 3 control replicates. Condition is fully confounded with study of origin: no batch spans more than one condition. All differential expression testing was therefore restricted to within-condition comparisons (treatment vs. control within a single study), and cross-condition claims were made only at the level of gene-list and directional comparison, not through a shared statistical model. The information regarding dataset and sample ideas are provided in Table 1.

Condition	GEO accession	Control replicates	Treatment replicates
Hypoxia (day 0 hypoxia persistence v/s day 4 reactivation control)	GSE166622	SRR3725597, SRR3725598, SRR3725599	SRR3725585, SRR3725586, SRR3725587
Iron starvation (1week low iron v/s high iron control)	GSE67035	SRR1917703, SRR1917704, SRR1917705	SRR1917709, SRR1917710, SRR1917711
Rifampicin (rifampicin 0.02 ug/mL at 72h v/s no drug at 72h)	GSE83814	SRR13687749, SRR13687750, SRR13687751	SRR13687767, SRR13687768, SRR13687769

Table 1. Overview of the three independent public RNA-seq datasets used in this study, one per stress condition. Each condition contributes 3 control and 3 treatment replicates (18 samples total); condition is fully confounded with the dataset of origin.

2.2 Bulk-RNA sequencing analysis of the 18 samples

Raw sequencing data for the 18 samples across the specified experimental conditions were retrieved from the Gene Expression Omnibus (GEO) repository. Initial quality control of the raw FASTQ files was conducted using FastQC and aggregated using MultiQC. To remove low-quality bases and residual adapter sequences, reads were processed with Trimmomatic, followed by a second pass of FastQC and MultiQC to verify post-trimming read quality. The

high-quality trimmed reads were then aligned to the reference genome (NC_000962.3) using Bowtie2. Alignment efficiency and mapping statistics were evaluated using Samtools flagstat and visualized via MultiQC. To confirm library strandedness, the infer_experiment module (RSeQC) was applied, and genomic feature quantification was subsequently performed using featureCounts to construct the final gene expression matrix for downstream differential analysis.

2.3 Differential expression testing

For each condition, samples were subset to that condition's study, and the group factor (treatment/control) was relevelled so that the control condition served as the reference level, giving log₂ fold changes with a consistent treatment-vs-control interpretation across conditions. Low-count genes were removed using a replicate-size-aware rule, retaining genes with a count ≥ 10 . Differential expression was fit with DESeq2's negative binomial generalized linear model (design = ~group), and log₂ fold change estimates were shrunk using the apeglm estimator, the current DESeq2-recommended method for coefficient-based shrinkage. Significance was defined as Benjamini–Hochberg adjusted p-value (padj) < 0.05 combined with an effect-size threshold of $|\log_2 \text{ fold change}| > 1$, applied uniformly across all reporting and visualization steps.

2.4 Cross-condition comparison

Significant gene sets from each condition were compared using set intersection (genes significant in all three conditions) and set difference (genes significant in exactly one condition, excluded from the union of the other two), visualized as a three-way Venn diagram. Because set-level overlap alone does not indicate whether shared genes move in the same direction, a second analysis restricted to the three-way-shared gene set classified each gene as directionally concordant (same sign of log₂ fold change in all three conditions) or discordant (sign differs across conditions), on the reasoning that concordant genes are the more defensible candidates for a shared, condition-independent stress response. A heatmap of z-scored, variance-stabilized expression (recomputed per condition with blind = FALSE to match the DE model) was generated for the concordant gene set only, with per-condition matrices combined column-wise; this combination is valid only because each condition's samples are mutually exclusive, and does not constitute a batch-corrected joint expression space.

2.5 Pathway enrichment

KEGG pathway enrichment was performed per condition using the M. tuberculosis H37Rv KEGG organism code (mtu) via clusterProfiler::enrichKEGG, with the enrichment background restricted to genes actually tested in that condition (post low-count filtering), to avoid biasing

enrichment statistics with genes that were never eligible for detection. KEGG identifier formatting for the mtu organism is a known source of false-negative (zero-hit) results due to locus-tag case and formatting mismatches. Any condition returning no enriched pathways was flagged for manual verification of gene ID formatting before being interpreted as a true absence of enrichment.

3. Results

Raw FASTQ quality across the 18 samples was assessed with FastQC/MultiQC. All samples passed standard quality modules, and per-base sequence content, adapter content were addressed by Trimmomatic trimming. Post-trim reassessment confirmed resolution of these issues in samples. Bowtie2 alignment to the H37Rv reference (NC_000962.3) yielded a mean overall alignment rate of $72\% \pm 4\%$. RSeQC's infer_experiment confirmed reverse strandedness library preparation across all samples, consistent with the featureCounts strand setting used for quantification.

To evaluate global transcriptional variability across conditions, principal component analysis (PCA) was performed on the normalized bulk RNA-seq dataset comprising 18 samples across three stress conditions: hypoxia, iron starvation, and rifampicin treatment (Figure 1). The first two principal components accounted for 67% of the total variance across samples (PC1: 43%, PC2: 24%). Biological replicates within each condition demonstrated tight clustering, indicating strong intra-condition consistency and high sample quality (Figure 1A). Samples separated distinctively along PC1 and PC2 according to their functional stress state. However, because each stress condition originated from an independent dataset (GSE166622 for hypoxia, GSE67035 for iron starvation, and GSE83814 for rifampicin), study-level batch effects co-segregated completely with experimental condition (Figure 1B).

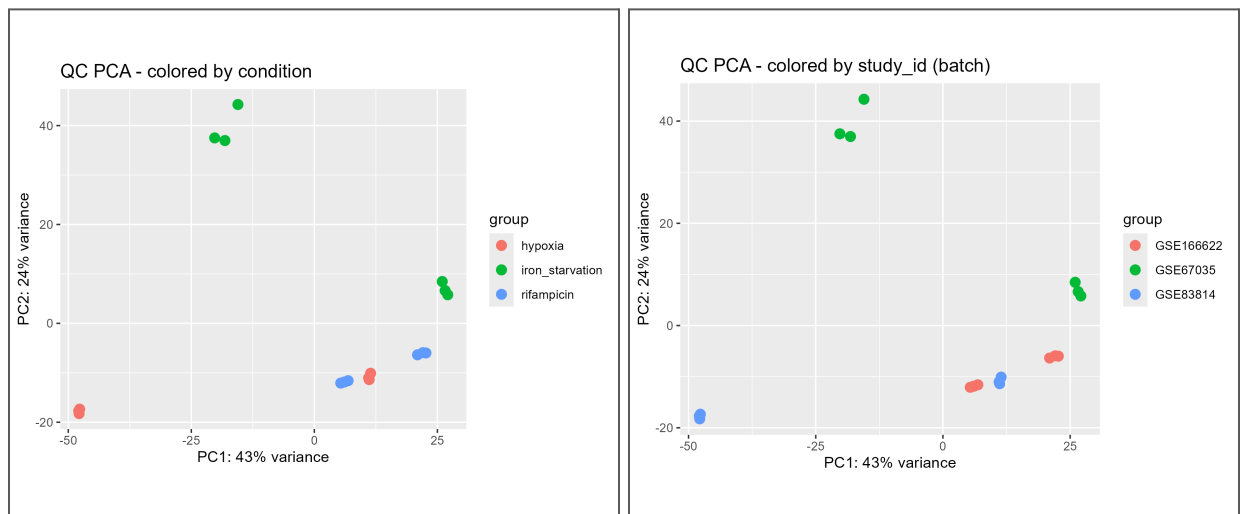


Figure 1. Principal component analysis of dataset based on functional state condition (A) and study Id (B).

To identify transcriptional signatures associated with distinct stress microenvironments, differential expression analysis (DESeq2) was performed across hypoxia, iron starvation, and rifampicin treatment relative to their respective control conditions.

Hypoxia: 1858 significant DEGs ($\text{padj} < 0.05$ & $|\log_2\text{FC}| > 1$) out of 3962 genes tested.

Rifampicin: 1501 significant DEGs ($\text{padj} < 0.05$ & $|\log_2\text{FC}| > 1$) out of 3916 genes tested.

Iron starvation: 1410 significant DEGs ($\text{padj} < 0.05$ & $|\log_2\text{FC}| > 1$) out of 3939 genes tested.

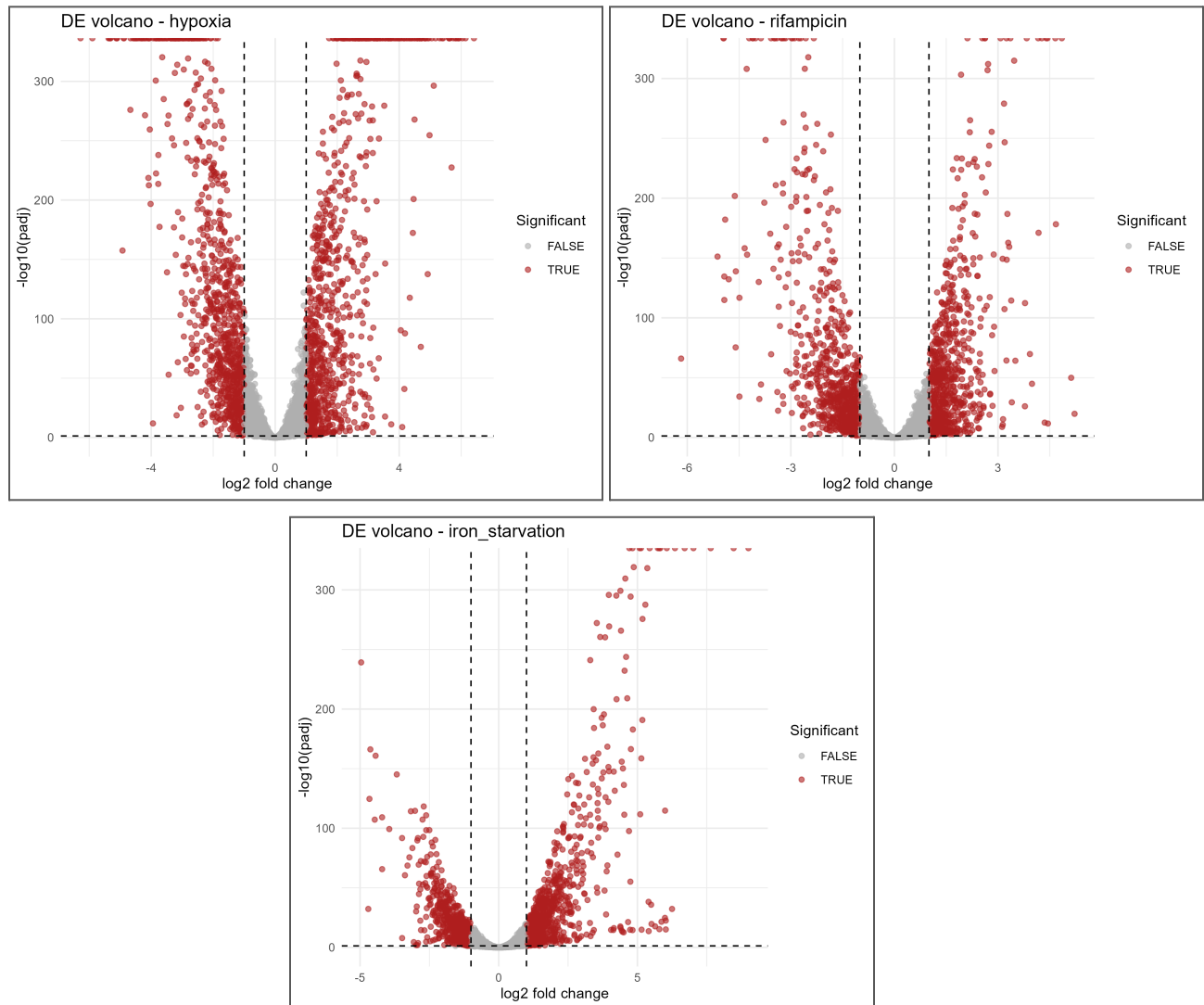


Figure 2. Volcano plots displaying global differential gene expression for *M. tuberculosis* exposed to (A) hypoxia, (B) rifampicin treatment, and (C) iron starvation. The x-axis represents

the $\log_2$ fold-change, and the y-axis indicates statistical significance as $-\log_{10}$ adjusted p-value. Vertical dashed lines mark the effect size threshold ($\log_2 FC = 1.0$), and the horizontal dashed line indicates the significance threshold ($p_{adj} = 0.05$). Statistically significant DEGs meeting both criteria are highlighted in red, while non-significant genes are shown in grey.

Intersecting the three conditions' significant gene sets identified 423 genes significant in all three conditions (three-way overlap, Fig. 3), 418 genes shared between hypoxia and rifampicin only, 439 genes shared between hypoxia and iron starvation only, and 201 shared between rifampicin and iron starvation only (Fig. 3). Condition-unique genes, significant in one condition and absent from the significant sets of the other two, numbered 578 for hypoxia, 459 for rifampicin, and 347 for iron starvation.

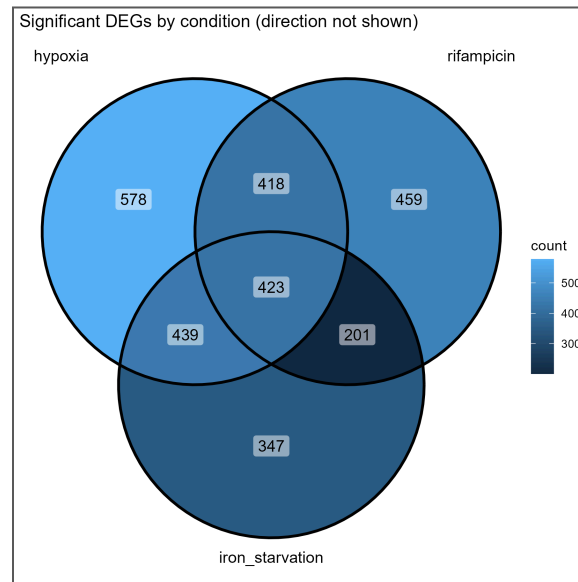


Figure 3. Venn diagram showing unique and shared significant DEGs ($|\log_2 FC| > 1.0$, $p_{adj} < 0.05$) across hypoxia, rifampicin, and iron starvation treatments. Numbers denote DEG counts per section, independent of regulation direction. Color bar indicates total gene counts.

Of 423 genes significant in all 3 conditions, 319 were concordant (same direction everywhere) and 104 discordant. Hierarchical clustering was performed on the set of shared differentially expressed genes (DEGs) that exhibited consistent directional responses across hypoxia, iron starvation, and rifampicin treatment (Figure 4). All control samples clustered together on the left branch, whereas all stress-treated samples clustered on the right branch, demonstrating that general stress perturbation exerts a stronger global effect on expression than study-specific baseline variations. Gene expression values (z-score normalized across rows) partitioned into two predominant transcription blocks. A large upper gene cluster demonstrated consistent upregulation across stress-treated conditions (red intensity) relative to control conditions (blue intensity), representing a core stress response network in *M. tuberculosis*. A lower gene cluster showed uniform downregulation under stress, particularly in hypoxia and iron starvation

conditions, indicative of conserved metabolic downregulation or growth suppression under non-permissive growth environments.

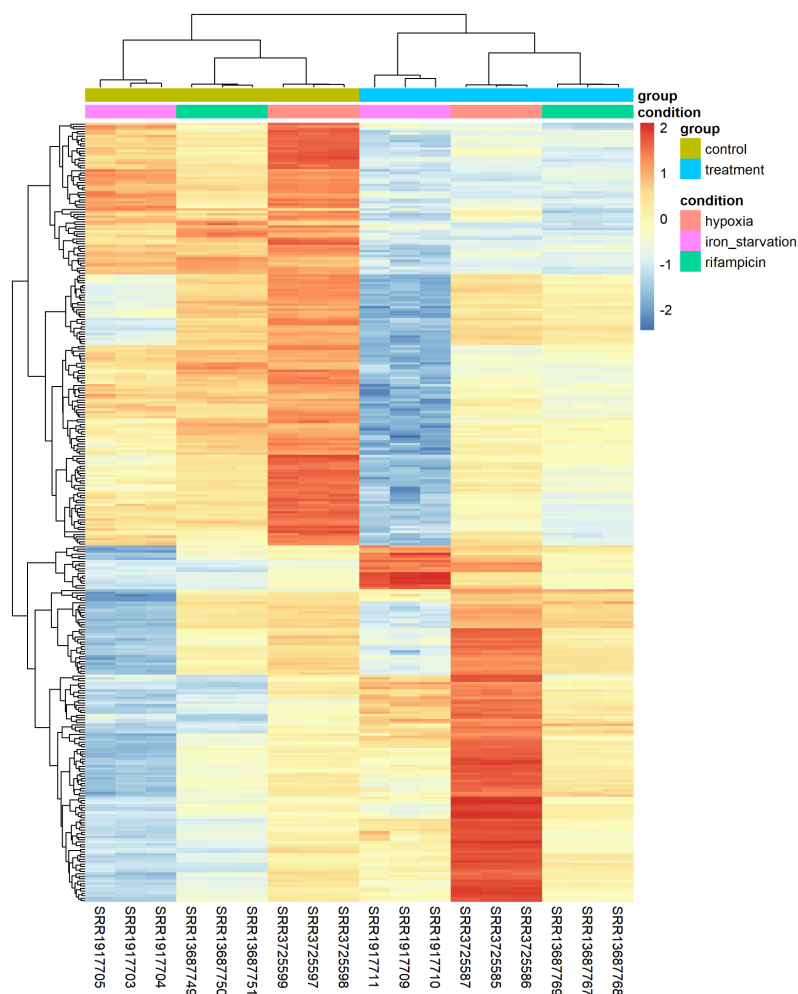


Figure 4. Hierarchical clustering of shared concordant DEGs across stress conditions. Heatmap displaying z-score normalized expression levels of shared differentially expressed genes (rows) across control and treatment samples (columns) for hypoxia, iron starvation, and rifampicin exposure. Both genes and samples were clustered using unsupervised hierarchical clustering with Euclidean distance and complete linkage. Sample annotation bars at the top denote experimental group (control vs. treatment) and stress condition (hypoxia, iron starvation, rifampicin). Relative expression levels are indicated by the color scale bar, ranging from blue (lower expression, z-score ≤ -2) to red (higher expression, z-score ≥ 2).

Gene	Mean log2FC	Known function / role	Confidence
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Rv3289c	4.79	Possible transmembrane protein, function otherwise uncharacterized; co-transcribed in the lat operon	High (operon), low (specific function)
Rv3290c (lat)	4.44	Lysine ϵ -aminotransferase converts L-lysine to L-pipecolate; one of the most strongly nutrient-starvation-induced genes in Mtb, regulated by LrpA	High
Rv2662	4.40	Not confirmed in this pass annotated locus, function not verified here	Not verified
Rv1158c	-4.33	Function uncharacterized; upstream of a TrcR/MtrA (stress two-component system) regulated operon	Low
Rv1884c	-4.22	Not confirmed in this pass	Not verified
Rv2780 (ald)	4.03	L-alanine dehydrogenase (Ald) hydrolyzes L-alanine, dampens host antimicrobial peptide induction, promotes intracellular survival; a well-known dormancy/hypoxia-associated gene	High
Rv1435c	-3.99	Not confirmed in this pass	Not verified
Rv1157c	-3.98	Function uncharacterized; upstream of TrcR/MtrA-regulated operon (paired with Rv1158c)	Low
Rv3477	-3.85	Not confirmed in this pass	Not verified
Rv1754c	-3.83	Function uncharacterized; a direct MtrA-regulon binding target	Low

Table 2. Top 10 Shared, Concordant Genes (significant + same direction in all 3 conditions): (Ranked by mean $|\log_2FC|$ across hypoxia, rifampicin, iron starvation.)

The genes uniquely significant to iron starvation converge strongly on a coherent, expected biological program. Five of the top ten (Rv2380c/mbtE, Rv2383c/mbtB, Rv2384/mbtA, Rv2386c/mbtI) belong to the mbt mycobactin siderophore biosynthesis cluster, and a sixth (Rv1466) belongs to the suf Fe-S cluster biogenesis operon, together indicating that the dominant iron-starvation-specific response is siderophore-mediated iron acquisition coupled with protection of iron-sulfur-dependent enzymes, consistent with well-established Mtb iron biology and lending internal validity to the DE pipeline's output.

The rifampicin-unique set is comparatively less characterized, but its two clearest hits, the Rv0196-Rv0197 operon (a putative transcriptional regulator paired with a possible oxidoreductase) and Rv3086/adhD (alcohol dehydrogenase) with its DosR-regulon-associated

neighbor Rv3087 point toward a redox- and transcription-remodeling response, plausibly reflecting the metabolic and respiratory stress imposed by RNA polymerase inhibition.

The hypoxia-unique genes are the least individually characterized of the three sets in current literature, but the presence of the *esxR/esxS* (Rv3019c/Rv3020c) ESX-secreted protein pair suggests a contribution from altered secretion/antigen presentation alongside the condition's core, better-known DosR-regulon response captured elsewhere in the dataset. Overall, each condition's unique gene set reflects a functionally distinct survival strategy, iron scavenging and Fe-S cluster protection for iron starvation, redox/transcriptional adjustment for rifampicin, and secretion-associated remodeling for hypoxia though a substantial fraction of genes in all three lists remain functionally unannotated, underscoring how much of the *Mtb* stress-response genome is still uncharacterized even for genes with strong, reproducible expression signals.

Top 10 genes unique to Hypoxia		Top 10 genes unique to Rifampicin stress response		Top 10 genes unique to iron starvation response	
Gene	Mean log2FC	Gene	Mean log2FC	Gene	Mean log2FC
Rv1895	4.95	Rv2812	5.22	Rv3839	8.47
Rv3019c (<i>esxR</i>)	4.92	Rv2813	4.35	Rv2383c (<i>mbtB</i>)	7.64
Rv3020c (<i>esxS</i>)	4.70	Rv0197	4.27	Rv2386c (<i>mbtI</i>)	6.70
Rv3833	4.66	Rv0196	4.17	Rv2122c (<i>hisE</i>)	6.00
Rv3167c	4.49	Rv1575	3.78	Rv3402c	5.82
Rv1875	4.40	Rv3087	-3.77	Rv2380c (<i>mbtE</i>)	5.44
Rv3463	4.37	Rv3086 (<i>adhD</i>)	-3.73	Rv2384 (<i>mbtA</i>)	5.18
Rv2660c	4.10	Rv3169	-3.57	Rv1344	5.10
Rv0519c	-4.09	Rv0655	-3.46	Rv2621c	4.82
Rv0140	4.00	Rv2935	3.36	Rv1466	4.15

Table 3. List of top 10 genes for each functional state based on their mean log2FC values.

4. Discussion

The comparative transcriptomic framework applied here identified both shared and condition-specific components of the *M. tuberculosis* stress response across hypoxia, rifampicin exposure, and iron starvation. Each condition induced a substantial fraction of the transcriptome (1858, 1501, and 1410 significant DEGs for hypoxia, rifampicin, and iron starvation, respectively), consistent with the scale of reprogramming previously reported for these stresses individually. The existence of 423 genes significant across all three conditions and more specifically, 319 genes that were both significant and concordant in direction suggests a core set of genes recruited into a general stress- or persistence-associated transcriptional program regardless of the specific triggering stimulus, alongside a larger set of condition-unique genes (578, 459, and 347 for hypoxia, rifampicin, and iron starvation, respectively) that likely reflect stimulus-specific adaptations.

The enrichment of oxidative phosphorylation genes among the hypoxia DEG set is consistent with the known remodeling of the *M. tuberculosis* respiratory chain under oxygen limitation. It should be noted that KEGG over-representation testing, as applied here, identifies pathways whose member genes are disproportionately represented among the significant DEGs without distinguishing whether those genes are up- or down-regulated; a direction-stratified enrichment (testing up- and down-regulated genes separately) would be needed to determine whether this reflects a coordinated shutdown, activation, or reorganization of the pathway, and is noted as a natural extension of this analysis.

5. Limitations

Condition and study of origin are fully confounded (one dataset per condition), so no statistical model in this analysis can separate a true biological effect of condition from a technical batch effect of study; all cross-condition claims are made at the level of gene lists and direction, not through a shared statistical model.

Each condition is supported by only 3 treatment and 3 control replicates from a single study, limiting statistical power and the generalizability of any single-gene finding; results should be treated as hypothesis-generating rather than definitive.

The combined expression matrix used for the heatmap is a visual convenience only: each condition's samples were variance-stabilized independently and combined column-wise because sample sets are mutually exclusive across conditions; this does not constitute a batch-corrected joint expression space, and absolute color intensities are not directly comparable in a cross-condition mean-centered sense.

6. Future Prospects

Beyond pathway-level and knockout-based follow-up, the differential expression signatures generated here could seed a machine learning classifier capable of predicting *Mtb*'s physiological/stress state directly from expression profiles, effectively learning a

condition-specific "transcriptional fingerprint" for hypoxia, rifampicin exposure, and iron starvation. With only 18 samples across three conditions, however, the current dataset is far too small to train a robust model directly; the more realistic near-term step is to use this analysis to define a compact, biologically-motivated feature set (e.g., the top concordant shared genes, the *mbt/suf* iron-response signature, and DosR-regulon members) that could then be tested against larger public Mtb expression compendia (GEO, ArrayExpress) using simple, interpretable models such as logistic regression, random forest, or elastic-net classifiers, approaches well suited to small-*n*, high-dimensional transcriptomic data and less prone to overfitting than deep learning here. Such a classifier would have two practical uses: first, as a computational tool to infer the likely physiological state of Mtb in ambiguous samples (e.g., clinical sputum transcriptomes, macrophage-internalized bacteria) where the true condition is unknown; second, as a hypothesis-generating step, since feature importance rankings (e.g., SHAP values) could surface additional condition-discriminating genes beyond the top fold-change hits used here, some of which may be low-fold-change but highly stable markers not captured by a $\text{padj}/\log_2\text{FC}$ threshold alone. Any such effort should be treated as exploratory rather than confirmatory until validated on an independent, ideally multi-strain and multi-lab dataset, given how easily small transcriptomic training sets overfit to batch-specific rather than biology-specific signal, a caution that applies with particular force here, given this project's own condition/study confound.

7. References

Bustad E, Petry E, Gu O, Griebel BT, Rustad TR, Sherman DR, Yang JH, Ma S. Predicting bacterial fitness in Mycobacterium tuberculosis with transcriptional regulatory network-informed interpretable machine learning. bioRxiv [Preprint]. 2024 Sep 25:2024.09.23.614645. doi: 10.1101/2024.09.23.614645. Update in: Front Tuberc. 2025;3:1500899. doi: 10.3389/ftubr.2025.1500899. PMID: 39386570; PMCID: PMC11463588.

Immanuel SRC, Arrieta-Ortiz ML, Ruiz RA, Pan M, Lopez Garcia de Lomana A, Peterson EJR, Baliga NS. Quantitative prediction of conditional vulnerabilities in regulatory and metabolic networks using PRIME. NPJ Syst Biol Appl. 2021 Dec 6;7(1):43. doi: 10.1038/s41540-021-00205-6. PMID: 34873198; PMCID: PMC8648758.

Yoo R, Rychel K, Poudel S, Al-Bulushi T, Yuan Y, Chauhan S, Lamoureux C, Palsson BO, Sastry A. Machine Learning of All Mycobacterium tuberculosis H37Rv RNA-seq Data Reveals a Structured Interplay between Metabolism, Stress Response, and Infection. mSphere. 2022 Apr 27;7(2):e0003322. doi: 10.1128/msphere.00033-22. Epub 2022 Mar 21. PMID: 35306876; PMCID: PMC9044949.

Shi C, Liu X, Chen D, Wang T, Wang Y, Cai N, Li Z, Hu Y, Cai Y, Chen X. Dual transcriptomic analysis unraveling the immune landscape and host-pathogen interactions during Mycobacterium

tuberculosis infection. *iScience*. 2025 Nov 17;28(12):114102. doi: 10.1016/j.isci.2025.114102. PMID: 41492386; PMCID: PMC12765389.

Ma S, Minch KJ, Rustad TR, Hobbs S, Zhou SL, Sherman DR, Price ND. Integrated Modeling of Gene Regulatory and Metabolic Networks in *Mycobacterium tuberculosis*. *PLoS Comput Biol*. 2015 Nov 30;11(11):e1004543. doi: 10.1371/journal.pcbi.1004543. PMID: 26618656; PMCID: PMC4664399.

Peterson EJR, Abidi AA, Arrieta-Ortiz ML, Aguilar B, Yurkovich JT, Kaur A, Pan M, Srinivas V, Shmulevich I, Baliga NS. Intricate Genetic Programs Controlling Dormancy in *Mycobacterium tuberculosis*. *Cell Rep*. 2020 Apr 28;31(4):107577. doi: 10.1016/j.celrep.2020.107577. Erratum in: *Cell Rep*. 2021 Jun 15;35(11):109287. doi: 10.1016/j.celrep.2021.109287. PMID: 32348771; PMCID: PMC7605849.

Matern WM, Rifat D, Bader JS, Karakousis PC. Gene Enrichment Analysis Reveals Major Regulators of *Mycobacterium tuberculosis* Gene Expression in Two Models of Antibiotic Tolerance. *Front Microbiol*. 2018 Apr 4;9:610. doi: 10.3389/fmicb.2018.00610. PMID: 29670589; PMCID: PMC5893760.

Du P, Sohaskey CD, Shi L. Transcriptional and Physiological Changes during *Mycobacterium tuberculosis* Reactivation from Non-replicating Persistence. *Front Microbiol*. 2016 Aug 31;7:1346. doi: 10.3389/fmicb.2016.01346. PMID: 27630619; PMCID: PMC5005354.

Sebastian J, Ma S, Rustad T, Bustad E, Thomas A, Singh S, Kumar P, Sherman D, Alland D. Divergent transcriptional regulation of redox-homeostasis and permeability modulate rifampicin tolerance and sensitivity in *Mycobacterium tuberculosis*. *Nat Commun*. 2025 Dec 14;17(1):459. doi: 10.1038/s41467-025-67152-2. PMID: 41392072; PMCID: PMC12800208.

Kurthkoti K, Amin H, Marakalala MJ, Ghanny S, Subbian S, Sakatos A, Livny J, Fortune SM, Berney M, Rodriguez GM. The Capacity of *Mycobacterium tuberculosis* To Survive Iron Starvation Might Enable It To Persist in Iron-Deprived Microenvironments of Human Granulomas. *mBio*. 2017 Aug 15;8(4):e01092-17. doi: 10.1128/mBio.01092-17. PMID: 28811344; PMCID: PMC5559634.